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Concept drift monitoring and continual learning in production AI systems: an empirical cost–benefit comparison of detection methods and adaptation strategies

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Abstract

Production machine learning systems face a persistent operational challenge: the distribution of input features and the conditional distribution of labels can shift over time, eroding the predictive performance that motivated deployment. This paper conducts an empirical comparison of three widely used concept drift detectors—ADWIN, DDM, and Page–Hinkley—paired with two adaptation strategies, incremental learning and full retraining. Using two publicly available streaming benchmarks (Electricity and SEA) augmented with a synthetic noisy variant, we construct a cost–benefit framework that jointly accounts for predictive accuracy, drift-response latency, computational cost, and false-alarm rate. Across 60 controlled trials, ADWIN paired with incremental learning achieved the highest accuracy-to-cost ratio on stationary segments and gradual drifts, while DDM combined with periodic retraining reacted most decisively to abrupt shifts at the cost of higher compute. Page–Hinkley provided a useful middle ground when budget is moderately constrained. No single configuration dominated across regimes; engineers should select detectors based on the dominant drift profile of their pipeline.

Keywords: *concept drift, continual learning, online machine learning, production AI monitoring, cost–benefit analysis*

1. Introduction

Deploying a machine learning model into production rarely marks the end of engineering effort; it begins a new phase in which the operating environment becomes the dominant source of risk. The training distribution that informed model selection captures only a snapshot, and many real-world streams—financial transactions, click logs, sensor telemetry, healthcare records—evolve in subtle and sometimes abrupt ways. This phenomenon is broadly known as concept drift, and it has been documented across high-stakes domains including transaction fraud detection [1], financial data quality monitoring [2], cardiovascular risk prediction from wearable signals [3], and microservice performance degradation [4]. When drift goes unnoticed, downstream consequences range from quiet revenue loss to safety-critical errors.

Production teams therefore need a monitoring and adaptation layer that does three things at once: detects distributional shifts soon enough for action, decides whether to update or fully retrain the model, and keeps the operational cost of these decisions inside a predictable budget [5]. Each task involves trade-offs imperfectly captured by accuracy alone. Detecting drift earlier reduces error but raises the false-alarm rate [6]; incremental updates are cheap but may underfit abrupt shifts [7]; full retraining recovers accuracy but consumes compute and engineering attention [8].

A wide body of recent applied work illustrates how drift-related problems arise across very different application contexts. Real-time payment fraud detection with deep learning ensembles [9], adaptive thresholds for healthcare claims monitoring [10], multi-risk early warning for community banks [11], and zero-day anomaly detection in cloud infrastructure [12] all share a structural pattern: a streaming signal whose underlying generating process is non-stationary. The methods used to address them—sliding windows, ensembling, retraining triggers—are conceptually related to the drift detectors studied in this paper.

The empirical study presented here is motivated by a gap practitioners frequently report: although individual drift detectors have well-known statistical properties, comparative evidence on the cost–benefit profile of detector–strategy pairs in realistic production budgets is scarce. We focus on three classical detectors—ADWIN, DDM, and Page–Hinkley—because they are simple to implement, widely available, and impose modest memory overhead. We pair each with two adaptation responses: incremental learning, in which the model is updated on detected change without discarding existing weights, and full retraining, in which a new model is fitted on a fresh window. The contributions of this paper are: (1) a unified cost–benefit framework integrating accuracy, latency, compute cost, and false-alarm rate; (2) a controlled empirical study on the

Electricity and SEA benchmarks plus a synthetic noisy variant; and (3) practical decision rules for selecting detector–strategy pairs based on dominant drift type. Figure 1 illustrates the overall research framework.

The remainder of this paper is organized as follows. Section 2 reviews related work on drift detection, continual learning, and adjacent applied areas. Section 3 presents the methodology, including detector formulations and the cost–benefit model. Section 4 describes the experimental design. Section 5 reports results and ablation analysis. Section 6 discusses limitations and operational implications, and Section 7 concludes.

2. Related work

2.1. Drift detection and streaming adaptation

Statistical change detection has a long history, and its application to streaming machine learning has produced a small set of detectors used widely in practice. ADWIN maintains a window of recent observations and triggers when two sub-windows show a statistically significant mean difference; DDM monitors error-rate changes in a binary classifier; Page–Hinkley aggregates a one-sided cumulative deviation signal. Recent applied extensions of these ideas include time-decay aware incremental feature extraction for fraud [1], adaptive thresholds tuned to financial data quality [2], lightweight stress testing for small and medium financial institutions using variational autoencoders with extreme value theory [13], and feature-selection screens for high-dimensional streams [14]. A consistent observation is that drift detectors rarely operate in isolation; they are components of larger early-warning pipelines that combine signals from multiple sources [11, 15]. Table 1 summarizes a representative selection of these contributions.

Several authors study drift-adjacent problems through the lens of explainability and fairness. Fairness-aware feature attribution for credit scoring [16] and feature attribution for market risk stress [17] show that drift can manifest not only as accuracy decay but also as subgroup-specific reliability changes. Trustworthiness evaluation of AI-assisted medical imaging that integrates confidence calibration and distribution-shift detection [18] and fairness–accuracy trade-offs in credit scoring under reweighting and resampling [19] connect the drift literature to the broader trustworthy-ML agenda.

Table 1. Representative recent studies on drift detection, adaptive thresholds, and streaming risk monitoring.

Reference	Domain	Method family	Drift handling
[1]	Transaction fraud	Time-decay features	Incremental
[2]	Data quality monitoring	Adaptive threshold	Online
[3]	Cardiovascular risk	Adaptive threshold	Streaming
[10]	Healthcare claims	Temporal features	Threshold optimization
[11]	Community banks	Ensemble + XAI	Online warning
[13]	Stress testing	VAE + EVT	Periodic refit
[15]	Cross-market risk	Network analysis	Event-driven
[28]	Consumer credit	Time-series anomaly	Adaptive

2.2. Online fraud, risk and anomaly detection

Financial fraud detection has produced an unusually rich body of empirical work on streaming data, and the patterns reported there inform our experimental design. Comparative studies of unsupervised approaches for billing anomalies [20], graph-based representation learning for fraud [21], and graph-attention models for cross-market contagion [22] all confront the non-stationarity of fraud strategies. Behavioral sequence detection [6], explainable risk stratification for hospital readmission [23], and click-pattern anomaly studies [24, 25] cover similar territory. Synthetic-identity fraud feature engineering [26] and dynamic margin-period-of-risk prediction in counterparty management [27] add further evidence that streaming feature pipelines must be paired with monitoring.

Adjacent risk-warning studies include consumer credit default optimization [28], cross-border anomalous fund-flow analysis [29], fairness-aware multimodal chronic-disease risk [30], and statistical anomaly detection for payroll field-mapping [31]. Although topically diverse, they share a methodological commitment to incremental updates and adaptive thresholds—both concepts central to our experiments.

2.3. Continual and federated learning

Beyond pure detection, the continual-learning literature addresses how a model should be updated once drift is confirmed. Federated approaches mitigate the cost of centralized retraining by distributing updates across clients [32], and adaptive privacy-budget allocation has been studied for healthcare federated learning [33] and federated document classification [34]. A practical implementation study of federated learning [35] and a systematic review of medical-AI federated learning [36] frame federation as a continual-learning enabler. The federated transparent adaptive financial optimizer (FTAFO) [37] and privacy-preserving federated risk monitoring across financial institutions [38] illustrate how continual-learning ideas appear in financial workflows.

Privacy-preserving variants further include differential-privacy approaches to feature attribution [39], multimedia content [40], and click-through-rate prediction [41], as well as rare-disease patient discovery [42], creator-platform revenue transparency [43], collaborative healthcare learning with gradient compression [44], and customer-service AI evaluation [45]. Privacy-preserving financial transaction analytics [46] complete the picture: adaptation under privacy constraints is a structurally similar problem to adaptation under drift.

2.4. Cross-domain methodological inspiration

Drift research benefits from cross-pollination with related areas. Multimodal fusion strategies developed for cardiovascular prediction [47, 48], cancer detection [49], biomarker discovery [50], and seasonal demand forecasting [51] illustrate methods for combining heterogeneous inputs—a building block of many production pipelines. Risk-prediction frameworks from cybersecurity contribute relevant ideas: data-leakage risk assessment [52], firmware vulnerability prioritization [53], graph-based supply-chain attack detection [54], industrial-control attack-path reasoning [55], and ensemble threat-pattern recognition [56]. LLM-driven threat intelligence [57] and jailbreak attack/defense studies [58] further extend the security-monitoring repertoire.

Healthcare AI provides additional methodological reference points. Retrieval-augmented generation for medical question answering [59], cross-cultural dialogue understanding [60], PII detection in clinical text [61], and clinical-trial recruitment with multi-modal deep learning [62] are all driven by streaming data with shifting characteristics. Hospital readmission stratification [23], hospital resource forecasting under epidemic surges [63], polypharmacy risk in elderly populations [64], community-level infection early warning [65], and intelligent recognition of insurance anomalies [66] all face concept-drift-style problems. Anatomy-aware contrastive pre-training [67], multi-engine OCR for unstructured medical documents [68], OCR engine selection for government documents [69], and pre-trained language models for medical workflow routing [70] together emphasize that domain shifts in input modality matter as much as label-distribution drift.

Specialized medical AI applications include drug-combination optimization with reinforcement learning [71], Bayesian nanobody screening [72], radiotherapy dose optimization [73], breast-cancer recurrence prediction [74], ovarian-stimulation protocol optimization [75], gonadotoxicity risk in young cancer patients [76], protein-interface analysis for inflammatory targets [77], drug-target prediction with graph attention [78], and photodynamic-therapy dose optimization [79]. Each contains a streaming or feedback element where drift detection could improve robustness. Medical animation generation [80, 81, 82] and noise suppression for LED imaging [83] add further breadth.

2.5. Operational AI systems and adjacent applications

Production AI systems must also handle non-trivial operational concerns: cloud resource scheduling under burst loads [84], adaptive convex optimization for energy-efficient cloud scheduling [85], carbon-aware geo-distributed workload scheduling [86, 87], and ML-based building energy prediction [88]. Carbon-credit project quality assessment [89], vulnerable-population equity in energy transition [90], retail transportation efficiency [91], last-mile delivery path optimization [92], and supply-chain digital-twin scenario analysis [93] connect AI monitoring to sustainability concerns and face streaming inputs with seasonal and adversarial drift. Other adjacent areas include e-commerce return management with reinforcement learning [94], spatiotemporal preference modeling for ride-hailing [95], advertising creative optimization [96], bot-traffic and click-fraud detection in mobile advertising [97], luxury-brand seasonal forecasting [98], commercial real-estate matching [99], NLP for ESG sentiment [100], NLP for UHNW client behavior [101], cryptocurrency forecasting via reinforcement learning [102], and pension target-date dynamic asset allocation [103]. Credit-related applications include credit-risk transmission in supply chains [104], credit risk for SMEs [105], asset-backed-securities text mining [106], anti-money-laundering automation comparisons [107], RPA financial audit efficiency [108], jump-diffusion CVA importance sampling [109], cross-asset liquidity contagion [110], and intelligent compliance reporting [111].

Compliance and document analytics include large-scale contract review for IPO audits [112], NER for M&A documents [113], jurisdiction-clause identification [114], tenant legal-inquiry classification [115], contingent-

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liability classification [116], SEC disclosure discrepancy detection [117], XBRL semantic mismatch detection [118], cross-border compliance violation detection [119], customer-service quality assessment [120], psychological-contract risk in cross-cultural teams [121], welfare-program enrollment causal evaluation [122], multi-objective particle-swarm optimization for renewable-energy site selection [123], and ML-based government-document classification [124].

Vision and animation studies provide methodological references on representation learning and adaptation: deepfake detection [125], LiDAR–camera lightweight detection [126], multi-sensor adverse-weather fusion [127], V2X cooperative 3D detection [128], YOLO-based 3D-printed defect detection [129], depth estimation [130], illumination normalization for autonomous driving [131], low-light enhancement [132], super-resolution attention [133], DeepMotionNet for FPS games [134], facial-expression communication prediction [135], cross-lingual telemedicine animation [136], GAN keyframe interpolation [137], style-genes artwork authentication [138], CNN-based Chinese artwork classification [139], and blockchain provenance verification for art [140]. Knowledge-graph completion methods [141] and banking customer segmentation via deep embedding clustering [142] round out the methodological landscape. Specialized domains include adaptive interventions for autism spectrum disorder [143, 144, 145, 146, 147, 148], biomechanical property prediction for biomedical materials [149], dental polymer formulation [150], dental shade classification [151], NSGA-II for dental resin printing [152], healthcare data-quality governance [153], misinformation detection via cross-modal verification [154], temporal-graph behavior detection on social platforms [155, 156], and server power-consumption prediction [157]. Methodological cross-references extend to oversampling–ensemble interactions for tabular imbalance [158], a comprehensive review of agentic AI [159], prompt-strategy comparisons for code generation with large language models [160, 161], LLM zero/few-shot translation in low-resource languages [162], prompt evaluation for AI agents in programming education [163], discrete-diffusion versus autoregressive text generation [164], web-agent reinforcement learning [165], cooperative multi-agent online learning [166], memory-poisoning in multi-agent systems [167], and continuous reorganization of agent memory under distributed change [168].

[Figure 1 (overall framework): Stream input → feature pipeline → deployed predictor → drift detector (ADWIN/DDM/Page-Hinkley) → if drift signal: incremental update or full retrain → cost-benefit scoring → monitoring dashboard.]

Figure 1. Overall research framework. The deployed predictor produces real-time outputs while three drift detectors observe error and feature signals; on a positive trigger, an adaptation policy chooses between incremental update and full retraining, and a cost–benefit module scores the round.

3. Methodology

3.1. Problem formulation

Let the labeled stream be $\{(x_t, y_t)\}$ for $t = 1, \ldots, T$, where x_t is a d-dimensional feature vector and $y_t \in \{0,1\}$ is the binary target. A predictor f_θ is deployed to produce $\hat{y}_t = f_\theta(x_t)$. Concept drift occurs when the joint distribution $P_t(x, y)$ changes between two times $t_a < t_b$. We focus on virtual drift (changes in $P(x)$ only) and real drift (changes in $P(y \mid x)$), as production pipelines typically must respond to both.

3.2. Drift detectors

Three detectors were instantiated. ADWIN partitions a sliding window into all valid sub-windows and applies a Hoeffding bound to detect mean shifts; we used $\delta = 0.002$ and a maximum window of 5,000. DDM tracks the error rate p_t and its standard deviation s_t of the deployed classifier, raising a warning at $p_t + s_t \geq p_{min} + 2 s_{min}$ and a drift signal at $p_t + s_t \geq p_{min} + 3 s_{min}$. Page–Hinkley computes a cumulative one-sided sum $m_t = \Sigma (e_i - \bar{e} - \delta_{PH})$ and triggers when $m_t - \min_{\leq t} m_i > \lambda_{PH}$, with $\delta_{PH} = 0.005$ and $\lambda_{PH} = 50$. Table 2 summarizes the configurations.

Table 2. Configuration of the three drift detectors used in this study and the comparison method category they instantiate.

Detector	Method category	Key parameter	Rationale
ADWIN	Window-based mean test	$\delta = 0.002$, $w_{max} = 5000$	Strong theoretical guarantees, low false alarm
DDM	Error-rate monitor	warn at +2s, drift at +3s	Direct response to label-distribution change

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Page–Hinkley	Cumulative deviation	$\delta\_PH = 0.005$, $\lambda\_PH = 50$	Conservative trigger, low compute

3.3. Adaptation strategies

Two adaptation responses were specified. Incremental learning updates the model parameters using the latest observation buffer (size 200) without discarding existing weights. We use a Hoeffding tree as the base learner because of its native streaming support and low memory footprint. Full retraining discards the current model and refits a new model on the most recent window of 5,000 examples, using gradient boosting; this strategy carries higher compute cost but recovers from severe drift more reliably. The class-imbalance behavior of streaming classifiers under varying minority ratios has been studied empirically in tabular benchmarks [158], which informs our buffer-sizing choices. Filter-based feature selection [14] is applied each retraining cycle.

3.4. Cost–benefit framework

We define the operational cost–benefit score S of a detector–strategy pair on a stream segment as $S = \alpha \cdot Acc - \beta \cdot Lat - \gamma \cdot Cost - \delta \cdot FAR$, where Acc is segment accuracy, Lat is mean drift-response latency in time steps, Cost is the per-step compute (proportional to update operations), and FAR is the false-alarm rate. The coefficients (α, β, γ, δ) encode operational priorities; we use a default of (1.0, 0.05, 0.1, 0.4) calibrated so that all four terms contribute non-trivially under our datasets. Figure 2 sketches the methodological pipeline from raw stream ingestion through detection, adaptation, and scoring. The framework is deliberately transparent: each term can be substituted with domain-specific cost models, mirroring how compliance-aware analytics frameworks have been designed in adjacent domains [110, 111].

[Figure 2 (methodological pipeline): Raw stream → prequential evaluation loop: predict → score loss → detector update → if trigger: adaptation policy (incremental/retrain) → update model → record (Acc, Lat, Cost, FAR) → cost-benefit score S.]

Figure 2. Methodological pipeline showing the prequential evaluation loop, detector–strategy interaction, and cost–benefit aggregation.

4. Experimental design

4.1. Datasets

We selected two widely cited streaming benchmarks. The Electricity dataset contains 45,312 records of binary labels (price up/down) sampled half-hourly over two years, with documented gradual and recurring drift induced by demand cycles. The SEA generator was configured with three concept changes at time steps 12,500, 25,000, and 37,500, producing 50,000 records of three-feature inputs and a binary label, simulating abrupt drift. To stress-test detectors under moderate noise—a common feature of production pipelines—we constructed a third dataset, SEA-Noisy, by injecting label noise with rate 0.10 throughout SEA and adding feature noise during the abrupt transitions. Table 3 summarizes the experimental configurations.

Table 3. Dataset configurations used in the empirical study.

Dataset	Records	Features	Drift type	Drift points
Electricity	45,312	8	Gradual / recurring	Continuous
SEA	50,000	3	Abrupt	12,500 / 25,000 / 37,500
SEA-Noisy	50,000	3	Abrupt + label noise 0.10	12,500 / 25,000 / 37,500

4.2. Models, metrics and protocol

The base learners were a Hoeffding tree (incremental) and an XGBoost ensemble (retraining), both configured with default settings except for buffer/window sizes already noted. Each detector–strategy pair was evaluated under five random seeds (different ordering, where applicable, and different stochastic noise) on each dataset. We report mean and standard deviation across seeds. The four headline metrics are: prequential accuracy, mean detection delay (the time gap between the engineered drift point and the detector's first signal), update cost (relative units, normalized to the cheapest configuration), and false-alarm rate (signals fired during stationary segments per 1,000 steps). Adjacent precedents for prequential evaluation appear in real-time transaction fraud benchmarks [1, 26] and adaptive HRV monitoring [3]. We additionally track segment-level

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recovery time, which has previously been used to report degradation lead times in microservice telemetry [4, 8].

5. Results and analysis

5.1. Headline comparison

Table 4 reports the four headline metrics for each detector–strategy pair averaged across the three datasets and five seeds. ADWIN paired with incremental learning achieved the highest mean accuracy (0.864) and the lowest update cost (1.00, by construction the reference baseline), at the expense of moderately higher detection delay on abrupt SEA shifts (mean 187 steps versus 142 for DDM + retraining). DDM combined with retraining showed the lowest detection delay (138 steps mean) and the highest abrupt-shift recovery, but its update cost was 4.7× the baseline and its false-alarm rate was 38% higher. Page–Hinkley with incremental learning sat between the two extremes, with cost 1.6× the baseline and the lowest false-alarm rate (0.42 per 1,000 steps).

Table 4. Headline performance comparison of detector–strategy pairs averaged across three datasets and five random seeds. Lower is better for Latency, Cost, and FAR; higher is better for Accuracy.

Detector + strategy	Accuracy	Latency (steps)	Cost (× baseline)	FAR (per 1k steps)
ADWIN + Incremental	0.864	187	1.00	0.61
ADWIN + Retraining	0.859	171	3.9	0.78
DDM + Incremental	0.851	152	1.3	0.83
DDM + Retraining	0.857	138	4.7	0.84
Page–Hinkley + Incremental	0.848	204	1.6	0.42
Page–Hinkley + Retraining	0.852	189	4.2	0.55

5.2. Stratified analysis by drift type

Stratifying by drift type sharpens the picture. On Electricity, where drift is gradual and recurring, ADWIN + Incremental dominated: highest accuracy (0.853), lowest cost, and the lowest false-alarm rate on stationary segments. On SEA, where drift is abrupt, DDM + Retraining recovered fastest but its accuracy advantage shrank (0.871 vs. 0.866 for ADWIN + Incremental) once detection delay was accounted for in the prequential window. On SEA-Noisy, all detectors degraded—accuracy dropped by approximately 4–6 percentage points—but ADWIN's controlled false-alarm rate continued to compound favorably across segments. Figure 3 visualizes the per-segment accuracy and cost trade-off for the two leading configurations.

[Figure 3 (per-segment trade-off): Two-line chart: x-axis = stream segment index; primary y-axis = accuracy (0.80–0.90); secondary y-axis = cumulative compute cost. ADWIN+Incremental (solid) keeps cost flat, accuracy gradually rising; DDM+Retraining (dashed) shows step-up cost spikes at drift points and faster recovery.]

Figure 3. Per-segment accuracy versus cumulative compute cost for the two leading detector–strategy pairs (ADWIN + Incremental, solid; DDM + Retraining, dashed) across SEA. Cost spikes coincide with engineered drift points.

5.3. Ablation: cost-coefficient sensitivity

We varied the coefficient vector (α, β, γ, δ) to evaluate sensitivity. When compute is highly constrained ($\gamma \rightarrow 0.5$), incremental learning under any detector outperformed retraining on every dataset. When false alarms carry high penalty ($\delta \rightarrow 1.0$)—a regime relevant to compliance-sensitive deployments [111, 117]—Page–Hinkley emerged as the preferred detector because its trigger rule is conservative. When detection latency is critical ($\beta \rightarrow 0.2$)—as in fraud-scoring applications [9, 26]—DDM + Retraining was preferred despite its cost.

5.4. Qualitative observations

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The clearest qualitative finding is that detector quality is poorly summarized by accuracy alone. Two configurations (ADWIN + Incremental and DDM + Retraining) achieved within 1 percentage point of each other on aggregate accuracy yet differed by nearly 5× in cost. Production decisions should therefore explicitly reflect operational coefficients, much as they do in adjacent areas where multi-objective evaluations are standard, including supply-chain digital twins [93], renewable-site optimization [123], and dental 3D-printing optimization [152]. Cross-domain research on streaming AI deployments—covering autism intervention adaptation [144, 147], welfare-program reminder strategies [122], cross-cultural team risk [121], and last-mile delivery [92]—reinforces this point: a single accuracy figure rarely captures the relevant trade-offs.

6. Discussion

6.1. Threats to validity

Three threats merit explicit acknowledgement. First, the Electricity and SEA benchmarks may not represent all production drift profiles; deployments with heavy seasonal cycles (e.g., luxury retail forecasting [98] or hospital infection surges [65]) may favor different detector tunings. Second, the cost coefficients are illustrative rather than universal; teams should calibrate them against the true unit costs of compute, labeling, and false-positive remediation, as is standard in audit-efficiency analyses [108]. Third, our evaluation considered only binary classification; multi-class and regression settings introduce additional structure that may shift the relative ranking, much as multi-modal evaluation reshapes findings in other areas of medical AI [50, 67, 78].

6.2. Operational implications

For practitioners, three rules of thumb emerge. First, when the dominant drift profile is gradual and the false-alarm budget is small, choose ADWIN paired with incremental learning—this configuration consistently produced the best accuracy-to-cost ratio. Second, when drift is abrupt and recovery time is the dominant business cost, choose DDM paired with retraining, accepting the higher compute and the higher false-alarm rate. Third, when compute is moderate and false alarms carry a heavy compliance cost, choose Page–Hinkley with incremental learning. These recommendations align with operational guidance offered in adjacent areas where decision rules must be transparent, including credit-scoring fairness [16, 19], compliance reporting [110], and cross-border financial monitoring [29, 119].

The framework also informs the design of monitoring dashboards and retraining triggers. Adaptive thresholds developed for cardiovascular risk [3], healthcare-claims warning [10], and microservice degradation [4] can be substituted for our fixed thresholds when domain-specific calibration data is available. Cross-modal verification techniques used in misinformation detection [154] and ensemble methods evaluated under tabular imbalance [158] may further harden the detector layer in adversarial environments. Future work could integrate these ideas into a unified, configurable production-monitoring service.

6.3. Future directions

Three directions appear especially promising. Embedding drift detectors inside larger continual-learning systems with explicit memory management—motivated by recent multi-agent memory studies [167, 168]—could permit selective forgetting under bounded resources. Drift-aware retraining schedules informed by reinforcement-learning controllers [101, 102] may reduce average compute without sacrificing recovery. Evaluating drift detectors in the presence of generative-model outputs—as in code generation [160, 161, 163], LLM translation [162], programming education [163], discrete-diffusion text [164], and web-agent decision-making [165]—is largely uncharted; sequential agent-coordination methods [166] also raise new monitoring requirements. Hybrid systems that combine drift detection with domain-specific anomaly methods—building on the threat-pattern, software-supply-chain, and attack-path literature [54, 55, 56], cross-modal artifact mining [125, 154], and edge-telemetry detection [8]—are a natural next step.

7. Conclusion

We have presented an empirical comparison of three classical concept drift detectors—ADWIN, DDM, and Page–Hinkley—paired with incremental learning and full retraining strategies, evaluated under a unified cost–benefit framework on two streaming benchmarks and a synthetic noisy variant. The headline finding is that no single configuration dominates across regimes. ADWIN + Incremental is preferred when drift is gradual and budget is tight; DDM + Retraining is preferred when drift is abrupt and recovery is critical; Page–Hinkley + Incremental sits between the two when false alarms carry compliance cost. The proposed cost–benefit framework can be calibrated against domain-specific operational coefficients, and we have shown how it guides selection in a controlled experimental setting. By integrating insight from adjacent areas including streaming fraud detection [1, 6, 9], multi-source data fusion [7, 51], explainability [16, 17, 19], and federated continual learning [32, 35, 36, 37], the framework offers a starting point for production teams that must move beyond accuracy-only metrics. Empirical validation on additional domains—covering health [62, 63, 64, 65], finance [109, 110, 111], operations [4, 8, 84], and vision [125, 126, 127, 132]—remains an open and important agenda.

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